

Revanth Nagaraj Mallol

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SUMMARY

Data Engineer with 3+ years of experience designing, building, and operating scalable batch and real-time data pipelines across AWS, GCP, and Azure. Proven expertise in Python, SQL, PySpark, Apache Kafka, Airflow, and dbt with strong foundation in dimensional modeling, data quality engineering, and cloud data warehouse optimization. Experienced in healthcare data integration, HIPAA-compliant analytics, and patient-centric data solutions. Track record of delivering quantifiable business impact through cost reduction, latency improvements, and analytics platform modernization.

TECHNICAL SKILLS

Languages: Python, SQL, Bash, Java

Data Engineering: ETL/ELT, Distributed Pipelines, Dimensional Modeling (Kimball), SCD Type 2, Data Quality, Data Lineage, dbt, Data Governance

Big Data & Streaming: Apache Kafka, Apache Spark, PySpark, Apache Flink, Hadoop, Hive

Cloud Platforms: AWS (S3, Glue, EMR, Lambda, Redshift, Kinesis, RDS, IAM) | GCP (BigQuery, Dataflow, Dataproc, Pub/Sub) | Azure (ADF, Synapse, ADLS Gen2)

Healthcare Data: EHR/EMR integration, HIPAA compliance & governance, Claims processing, Patient analytics, HL7/FHIR standards awareness

Platforms & DevOps: Snowflake, Databricks, Delta Lake, Apache Iceberg, Airflow, Docker, Kubernetes, Terraform, GitHub Actions, CI/CD

BI & Monitoring: Great Expectations, Grafana, Metabase, Power BI, Tableau

Databases: PostgreSQL, MySQL, Snowflake, Amazon Redshift, BigQuery

EXPERIENCE

Data Engineer

Dec 2024 – May 2026

University of Texas at Arlington

- Architected real-time intent-data ingestion pipelines using Apache Kafka and Flink, processing 500M+ events/day to fuel AI-driven audience targeting; reduced model refresh SLA from 24 hrs to 4 hrs
- Designed Snowflake data warehouse schemas (Kimball star schema) for patient analytics and healthcare cost/quality measures; optimized query performance by 40% through clustering keys and materialized views
- Built and maintained 200+ dbt transformation models with 95%+ test coverage, orchestrated via Airflow DAGs; established CI/CD pipelines using GitHub Actions for automated schema validation and deployment
- Ensured data quality, integrity, security, and consistency across cloud platforms by applying systematic validation with Great Expectations and governance via dbt

Data Engineer

Jul 2021 – Jul 2024

L&T Finance Limited – NBFC & Financial Services

- Led development of a Python/Airflow ELT platform ingesting data from 6 heterogeneous source systems into a centralized PostgreSQL data warehouse serving analytics for 2M+ customers
- Designed and deployed an Apache Kafka streaming pipeline for event processing, reducing alert latency from 15 minutes to under 90 seconds – applicable to clinical alerting systems
- Authored 150+ dbt transformation models implementing SCD Type 2 for customer 360-degree views; integrated Great Expectations data quality checks, reducing downstream errors by 72%
- Migrated 8 legacy workflows to AWS Glue + S3 + Redshift, cutting monthly infrastructure costs by \$32K and improving pipeline SLA from 91% to 99.4%
- Implemented column-level lineage tracking eliminating data regression incidents by 65% – essential for audit trails and compliance in healthcare data systems
- Mentored 2 junior data engineers in Python best practices, Airflow DAG design patterns, and code review standards

Data Engineer – Internship

May 2025 – Aug 2025

UTA Campus Operations

- Designed and maintained MavExpress, a production Python ETL pipeline synchronizing student/staff access permissions across 6 data sources with real-time conflict resolution logic serving 40K+ users
- Built automated data ingestion frameworks integrating structured and semi-structured data from REST APIs into cloud platforms; improved pipeline throughput by 35%
- Implemented data quality checks, schema validation, and monitoring workflows; developed Power BI dashboards enabling cross-functional KPI monitoring

PROJECTS

StreamIQ – Real-Time LLM-Enriched Analytics Platform | *Kafka, PySpark, LangChain, Ollama, Airflow, dbt, MinIO, F*

- Architected end-to-end streaming platform ingesting 500M+ events/day via Kafka, enriching signals with LLM-powered analysis; achieved sub-second latency at 10K msg/sec – architecture applicable to real-time clinical data enrichment at healthcare scale
- Implemented Delta Lake for ACID-compliant storage, dbt semantic models for transformation, Airflow for orchestration, FastAPI + Grafana for real-time dashboards; full CI/CD via GitHub Actions

NYC Urban Analytics ELT Pipeline | *Airflow, dbt, PostgreSQL, PostGIS, Metabase, Python*

- Orchestrated daily ingestion of 50M+ rows/month into PostgreSQL; built Kimball-modeled dbt transformation layers with automated data quality tests and Metabase dashboards
- Implemented incremental Airflow DAG strategies reducing daily pipeline runtime by 45%; added Great Expectations checkpoints to catch schema drift and anomalies

EDUCATION

University of Texas at Arlington

Aug 2024 – May 2026

Master of Science, Data Science – GPA: 4.0/4.0

CERTIFICATIONS

Microsoft Certified Data Engineer (DP-700 – Microsoft Fabric) | Microsoft Certified Azure AI Fundamentals (DP-900) | GCP Data Engineer Professional (In Progress)

PUBLICATIONS & ARTICLES

“Generation of Realistic Image (Photo Stack)” | NeuroQuantology, Dec 2022, Vol. 20, Issue 22, pp. 1147–1155 | DOI: 10.48047/nq.2022.20.22.NQ10090

StackGAN-based text-to-image synthesis research; co-authored at Nitte Meenakshi Institute of Technology

“Introduction to Google Cloud Trace and Google Cloud Debugger” | GeeksforGeeks Cloud Computing, Mar 2023

Technical article on GCP observability tooling covering distributed tracing and runtime debugging workflows